

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Engineered and maintained scalable data pipelines using **AWS Glue (PySpark)**, processing over **2 TB** of data daily for downstream analytical workloads.
- Optimized **ClickHouse warehouse** performance through strategic indexing and query tuning, achieving sub-second query response times for critical back-office reporting.
- Implemented data contracts between back-office systems and the central data platform, ensuring tenant isolation via filter logic and reducing onboarding complexity to a configuration change.
- Designed and built the **feature-serving layer** for AI agents, delivering pre-computed features with **millisecond latency** by leveraging **Redis** and **FastAPI**.
- Integrated data ingestion from diverse sources including third-party **PostgreSQL** databases, **REST APIs** from affiliate platforms, and external **CRM** systems.
- Established comprehensive monitoring and alerting procedures for pipeline health, data freshness, and anomaly detection, including **Great Expectations** for automated validation.
- Managed CI/CD pipelines and infrastructure-as-code deployments using **Terraform** and **Docker** on **AWS ECS Fargate** for data workloads, ensuring automated testing and version-controlled deployments.

SKILLS

Cloud Platform – AWS Glue, PySpark, AWS ECS Fargate, S3, Lambda, CloudWatch, IAM, Terraform, CloudFormation, CDK

Data Warehousing & Databases – ClickHouse, PostgreSQL, MySQL, Oracle, Redis, DynamoDB

Data Pipelines & Orchestration – Apache Kafka, Amazon MSK, Spark Streaming, Apache Airflow, Apache NiFi, ETL, Data Contracts, Schema Governance

Programming Languages & Frameworks – Python, PySpark, SQL, Java, Spring Boot, REST APIs, FastAPI, Flask

Observability & Monitoring – Monitoring, Alerting, Anomaly Detection, Data Quality, Great Expectations, Splunk, Dynatrace

DevOps & CI/CD – CI/CD, Infrastructure-as-Code, Docker, Kubernetes, Helm, Git, Jenkins, Automated Testing

PROJECTS

Real-time Transaction Data Platform (HCL Technologies)

- Developed a real-time data ingestion pipeline using **Kafka** and **PySpark** to process financial transaction messages, handling approximately **1,500 events per second**.
- Engineered transformations and enrichments within the pipeline to support operational alerting and reconciliation workflows for a financial services client.
- Implemented robust error handling and monitoring using **Splunk**, ensuring high availability for critical transaction flow data.

Multi-Tenant Analytics Warehouse Design (MSc Project)

- Designed a normalized, multi-tenant data model in **PostgreSQL**, structuring schemas to support tenant isolation through a single filter layer, accommodating **5 distinct client types**.
- Developed **Power Query** scripts for ETL processes and performed data modeling to integrate disparate data sources for analytical reporting.
- Created interactive **Power BI** dashboards to visualize key performance indicators for business stakeholders, demonstrating data accessibility and insight generation.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)